

Aryaman Kaprekar

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SUMMARY

Cyber Physical Systems graduate with two years of combined research and industry experience in AI for safety-critical applications, spanning medical imaging and autonomous vehicles. Published peer-reviewed research in computer-aided diagnosis and led deployment of AI on embedded hardware for advanced driver assistance systems (ADAS).

EDUCATION

- Manipal Institute of Technology** Aug 2026
Bachelor of Technology (Hons.) in Cyber Physical Systems Manipal, India
(Minor Specialization: Computational Intelligence)
 - CGPA: 8.86/10.00
 - Relevant Coursework:* Machine Learning, Computer Vision, Unsupervised Intelligence in CPS, Soft Computing Techniques, Data Structures and Algorithms, Embedded Systems Programming

SKILLS

- Programming Languages:** Python, C++, C, Embedded C
- Frameworks & Tools:** PyTorch, ONNX, TensorRT, CUDA, OpenCV, Git, Linux, MATLAB, CMake, NumPy, Pandas, DVC, MLFlow, Docker
- Technical Domains:** Deep Learning, Computer Vision, Model Optimisation, Reinforcement Learning, Edge-AI, Digital Image Processing, IoT, Computer Architecture, Computer Networks

EXPERIENCE

- DeepGrid Semi Pvt Ltd** May 2025 - Jul 2026
AI Intern → ADAS Engineer (AI Team Lead) Hyderabad, India
 - Led the AI team across autonomous driving programs, driving research direction and technical decisions; co-designed a 28 nm autonomous driving SoC with embedded and RTL teams.
 - Built the full embedded deployment pipeline (PyTorch → ONNX → TensorRT → CUDA → C++) with multi-sensor fusion, raising hardware throughput $240\times$ ($0.05 \rightarrow 12$ FPS) and cutting model size by 86% (1.1 GB $\rightarrow$ 150 MB).
 - Re-architected the flagship end-to-end autonomous driving model (AD2) for edge hardware cutting compute cost 95% while achieving a Drive Score of 86 on Bench2Drive (Closed-Loop Evaluation).
- Advanced Healthcare Lab, MAHE** Dec 2024 - Jul 2026
Research Assistant Manipal, India
 - Reviewed and applied deep learning to computer-aided diagnosis across four clinical applications: brain tumors, ischemic stroke, left ventricular hypertrophy, and autism spectrum disorder.
 - Developed lightweight architectures, uncertainty quantification, and quantitative explainable AI across 5 modalities: 2D images, 3D volumes, phenotypic features, time-series data, and functional graphs.
 - Designed an automated brain tumor detection tool, whose novel explainability framework was validated by expert radiologists at 87.64% explanation correctness.

SELECTED ACADEMIC PROJECTS

- Road-Scene Segmentation (2025):** Conducted a systematic literature review on segmentation and built a hybrid GAN, reaching a competent 63% mIoU / 72% mPA on Cityscapes with only 11.3M params.
- Consumer Product Cost Prediction (2025):** Built a multimodal price prediction pipeline fusing ResNet50 and DistilBERT embeddings into an XGBoost regressor, attaining a SMAPE of 47.5.
- RL-based Batch Reactor Controller (2025):** Designed a hybrid RL controller (Policy Mirror Descent + TD learning with heuristic control terms) for a simulated batch reactor, achieving <0.1 °C steady-state tracking error.
- Generative Models for Reflection Removal (2024):** Built autoencoder/GAN pipelines for single-image reflection removal, yielding 89.77% mAP on a self-compiled dataset.

PUBLICATIONS

- (Author, *Nature: Scientific Reports*, 2026, Q1) *A multi-level attention CNN-transformer based framework for the detection of brain tumor using regional dual-score explainability.*
- (Co-Author, *Nature: Scientific Reports*, 2026, Q1) *TrustNet: A Lightweight Network with Integrated Uncertainty Quantification and Quantitative Explainable AI for Ischemic Stroke Detection in CT Images.*
- (Co-Author, *Elsevier: Informatics in Medicine Unlocked*, 2025, Q2) *Automated Identification of Left Ventricular Hypertrophy using Cardiac Ultrasound Imaging: A Systematic Review of AI-Driven Approaches.*

ACHIEVEMENTS AND EXTRA-CURRICULAR

- NPTEL:** Top 5% in India across multiple AI/Computer Vision NPTEL courses
- Qualcomm Vision X:** Finalist for the reflection-removal project at IIT-B.
- Workshops:** IoT/ML training at Bolt-IoT; Gesture Robotics at IIT-B.
- Other:** Elected Class Representative; NCC regimental training and social service; Everest Base Camp trek (5,364m).